Question: What is geometric rep. thy?

Answer: A good way to start arguments in GRT and.

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① Basic question: G reductive group over $\mathbb{C}$
Lie group

e.g. $G = GL_n(\mathbb{C})$
$SL_n(\mathbb{C})$
$Sp_{2n}(\mathbb{C})$
$\vdots$

What are $\mathrm{Irr}(G)$?

Ex: $G = GL_2(\mathbb{C})$ $\quad V = \mathbb{C}^2$

$$\mathrm{Irr}(GL_2) = \{(a,b) \in \mathbb{Z}^{\geq 0} \times \mathbb{Z}\}$$

$\mathrm{Sym}^a V \otimes \det^b \longleftarrow\!\shortmid (a,b)$

Strategy: pass to linearization

$G \rightsquigarrow \mathrm{Lie}(G) = \mathfrak{g} \overset{\text{def}}{:=} T_e(G)$

$m: G \times G \to G \rightsquigarrow$ extra structure $[-,-]: \mathfrak{g} \otimes \mathfrak{g} \to \mathfrak{g}$

② Classify $\mathrm{Irr}(\mathfrak{g})$.

Ex: $\mathfrak{g} = \mathfrak{sl}_2$ (trivial center) $\quad V = \mathbb{C}^2$

$\mathrm{Irr}(\mathfrak{sl}_2) \longleftarrow\!\shortmid \mathbb{Z}^{\geq 0}$ $\qquad \begin{pmatrix} 1 & * \\ & 1 \end{pmatrix} \subseteq GL_n$

$\mathrm{Sym}^n V \longleftarrow\!\shortmid n$

$T \subseteq G$

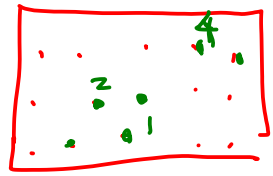
Outline: ① Fix Cartan/maximal torus $\mathfrak{h} \subseteq \mathfrak{g}$

$V \in \mathrm{Rep}(\mathfrak{g}) \rightsquigarrow V = \bigoplus_{\lambda \in \mathfrak{h}^*} V_\lambda$ restrict to $\mathfrak{h}$

V f.d. $\rightsquigarrow \lambda$ live in some lattice determined by $\mathfrak{g}$

$G = SL_3$ $\quad \mathfrak{g}^*_{\mathbb{R}}$

$T = \begin{pmatrix} * & & \\ & * & \\ & & * \end{pmatrix}$



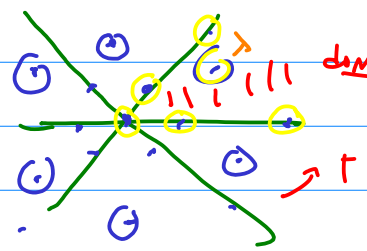
same lattice ??

$\mathfrak{g}$ 2 dim $\qquad V$

② $W = N(T)/T \curvearrowright V \in \mathrm{Rep}(G)$, preserve eigenvectors

acts on $\mathfrak{g}^*$ by reflections across hyperplanes

SL_3:



dominant

Fact: $\mathrm{Rep}(\mathfrak{g})$ finite

W-symmetric

③ Fix Borel subgroup, e.g. $B = \begin{pmatrix} * & * & * \\ & * & * \\ & & * \end{pmatrix}$

- Gives a notion of positivity
- "most positive chamber" (dominant)

Thm $\mathrm{Irr}_{\mathbb{R}}(\mathfrak{g}) \longleftrightarrow$ points lattice in dominant chamber

$L(\lambda) \longleftarrow \lambda$

Next question: What does $L(\lambda)$ look like?

i.e. what are dimensions of $\mathfrak{g}^*$-eigensp.?

Weyl character formula:

$$\sum_{\mu \in \mathfrak{g}^*} \dim(L(\lambda)_\mu) e^{\mu} = \frac{\sum_{w \in W} e^{w(\lambda+\rho)-\rho} \cdot (-1)^{\ell(w)}}{\prod_{\alpha \in \Delta^+} 1 - e^{-\alpha}}$$

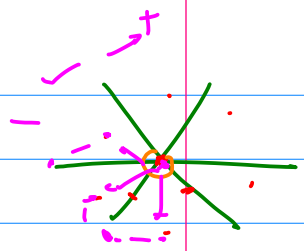
formal variable

"roots"

character of $\frac{1}{1-x}$

$= 1 + x + x^2 + \dots$

Interpretation: There is category of Rep($\mathfrak{g}$) called $\mathcal{O}$



contains some ∞-dim reps...

"standard modules" "Verma modules" $M(\lambda)$ always ∞-dim.

"freely generated" by $n^{\oplus}$

characters are polynomial

one

dominant

Formula comes from: formula is $\pi_1(\mathcal{O})$

W of those

Notation

$w \cdot \lambda$ " $w(\lambda+\rho)-\rho$

$$[L(\lambda)] = \sum_{w \in W} (-1)^{\ell(w)} [M(w(\lambda+\rho)-\rho)]$$

simples

coeff to matrix

n^- "free"

comes from a resolution of $L(\lambda)$ by $M(\lambda)$

BGG resolution

Question: Converse? Category $\mathcal{O}$ has finite length

- every object has a filtration
- subquotients i.e. "composition factors" unique.

every object is assembled from simples

W symm

$$[M(\lambda)] = \sum ?? \; L(\mu)$$

invert matrix

$L(\lambda)$ is f.d. iff λ dominant. Weyl char.

rot symm.

$L(\lambda)$ λ not dom no. Weyl char

Only have one row of matrix...

Question: Find character formula: λ dominant

$$[L(w \cdot \lambda)] = \sum_{\substack{y \leq w \\ \ell(y) \geq \ell(w)}} b_{y,w} [M(y \cdot \lambda)]$$

Kazhdan-Lusztig: thm $b_{y,w} = (-1)^{\ell} P_{y,w}(1)$

KL-polynomials
hard + complicated...

Beilinson-Bernstein
Brylinski-Kashiwara

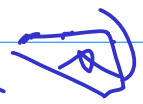
(Riemann-Hilbert)

Slogan: comes from geometry!

Warmup: (Borel-Weil-Bott)

Def: G/B flag variety (homogeneous space)

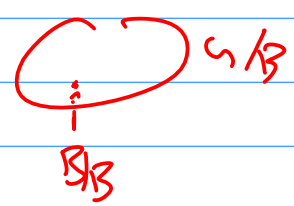
$G = GL_n(V)$ $\qquad G/B = \{ V_0 \subseteq V_1 \subseteq \dots \subseteq V_n \mid \dim(V_i) = i \}$

$B = ($  $)$

Exercise.

$\left(\begin{array}{c} G\text{-eq. lin bundles on} \\ G/B \end{array} \right) \longleftrightarrow \mathrm{Rep}(B)$

action on fiber at $B/B \in G/B$



$\Gamma(G/B, -) : Sh^G(G/B) \longrightarrow \mathrm{Rep}(G) : Sh^G(G)$

BWB: $\Gamma(G/B, \mathcal{L}(\lambda)) =$

inflate character λ of $H = B/[B,B]$ to U

$H^i(G/B, \mathcal{L}(\lambda)) = \begin{cases} V_{w\cdot\lambda} & \text{for } w\cdot\lambda \text{ dominant}, \ i = \ell(w) \\ 0 & \text{else.} \end{cases}$

i.e. shifts of irr. rep.

Ask: Is there a Lie algebra version?

$\mathfrak{g}$: invariant vector fields on G.

$G \curvearrowright X$, $\mathcal{F} \in Sh^G(X)$

$\{$ diff.

$\mathfrak{g} \to \mathrm{Vect}(X)$ $\qquad$ $\mathfrak{g} \curvearrowright \mathcal{F}$

Leads: definition of D-modules: quasicoh. sh. on X with action of "T_X"

$\boxed{\mathfrak{g}\text{-reps}}$

$\updownarrow$ Beilinson-Bernstein Γ.

$\boxed{\text{D-modules on } G/B}$ algebraic $\subset$

Riemann-Hilbert $\updownarrow$ solutions of diff eq

$\boxed{\text{constructible sheaves, perverse sheaves } (G/B)^{an}}$ analytic $\subset$

adding 'mixed' theory $\rightsquigarrow$ part of the conjecture.